GRADE 9 PRE-TECHNICAL STUDIES
CONTINUOUS ASSESSMENT TEST 1

TERM 1 | TIME: 1 HOUR 15 MINUTES | TOTAL: 50 MARKS

Name: _____ Adm. No.: _____

School: _____ Date: _____

INSTRUCTIONS

- Answer all questions.
- Draw neat, labelled sketches where required.
- Observe safety when describing workshop procedures.
- The marks for each question are shown in brackets.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. Define the term design brief. (2 marks)

_____

2. State TWO reasons for preparing a working drawing before making an artefact. (2 marks)

_____

3. Name TWO personal protective equipment (PPE) items used in a workshop. (2 marks)

_____

4. Differentiate between a measuring tool and a marking-out tool. (2 marks)

_____

5. State TWO properties of a suitable material for making a classroom stool. (2 marks)

_____

6. Give TWO examples of renewable energy sources. (2 marks)

_____

7. State TWO ways of reducing material wastage when making a product. (2 marks)

_____

8. What is meant by the term maintenance of tools? (2 marks)

_____

9. State TWO safe practices to observe when using a hand saw. (2 marks)

_____

10. Give TWO reasons for testing a finished product against its design brief. (2 marks)

_____

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Design process (8 marks)

(a) Explain the purpose of investigating a problem before designing a solution. (2 marks)

_____

(b) Outline FOUR stages that may be followed when developing a simple product from an identified need. (4 marks)

_____

(c) State TWO features of a good design brief. (2 marks)

_____

12. Materials and workshop practice (8 marks)

(a) Name TWO categories of materials commonly used in product making and give ONE example of each. (4 marks)

_____

(b) Explain TWO factors to consider when selecting a material for a product. (2 marks)

_____

(c) State TWO ways of keeping a workshop safe and orderly. (2 marks)

_____

13. Drawing and measurement (7 marks)

(a) State THREE pieces of information that should appear on a working drawing. (3 marks)

_____

(b) Explain why dimensions are important on a working drawing. (2 marks)

_____

(c) State TWO precautions to take when using a ruler or tape measure. (2 marks)

_____

14. Energy, tools and product evaluation (7 marks)

(a) Distinguish between renewable and non-renewable sources of energy. (2 marks)

_____

(b) State TWO ways of caring for hand tools after use. (2 marks)

_____

(c) Give THREE criteria that can be used to evaluate a finished product. (3 marks)

_____

MARKING SCHEME – GRADE 9 PRE-TECHNICAL STUDIES CAT 1

Total: 50 marks

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. A short written statement describing a problem or need to be addressed through design, including the purpose and relevant requirements. (2)

2. Any TWO: communicates shape and dimensions; guides accurate construction; identifies materials/components; helps estimate quantities; reduces errors and wastage. (1 each)

3. Any TWO: safety goggles; overalls/apron; gloves appropriate to the task; dust mask/respirator; safety boots; ear protection. (1 each)

4. A measuring tool is used to determine size or distance; a marking-out tool is used to transfer lines, points or shapes onto a material. (1 each)

5. Any TWO: strong; durable; stable; safe/non-toxic; affordable; readily available; easy to work; resistant to wear. (1 each)

6. Solar, wind, hydroelectric, geothermal, biomass or tidal energy. (1 each)

7. Any TWO: plan dimensions carefully; prepare a cutting list; use templates/nesting; measure accurately; reuse offcuts; select suitable stock sizes. (1 each)

8. Routine actions taken to keep tools in good working condition and safe to use, such as cleaning, drying, sharpening, lubricating and proper storage. (2)

9. Any TWO: secure the workpiece; keep hands away from the cutting line; use a sharp undamaged saw; cut steadily; keep the blade aligned; do not distract others. (1 each)

10. Any TWO: checks whether it meets user needs; confirms dimensions/function; identifies defects; checks safety, strength and finish; informs improvements. (1 each)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Design process

(a) It helps the designer understand the user's needs, causes and constraints so that the solution addresses the real problem. Award 1 mark per relevant point. (2)

(b) Any FOUR in a logical sequence, 1 mark each: identify/investigate the need; prepare a design brief/specifications; generate and sketch ideas; select/refine a solution; prepare working drawings; select materials/tools and plan making; make the product; test/evaluate and improve. (4)

(c) Any TWO: clearly states the problem/need; identifies intended user/purpose; gives key requirements or constraints; is specific and understandable. (1 each)

12. Materials and workshop practice

(a) Any TWO suitable categories with examples, 1 mark for category and 1 for example: metals – steel/aluminium; timber – pine/mvule; plastics – PVC/acrylic; textiles – cotton; ceramics – clay/tiles; composites – plywood/fibreglass. (4)

(b) Any TWO explained: strength/durability; cost; availability; ease of working; appearance; weight; safety; environmental impact; suitability for intended use. (1 each)

(c) Any TWO: keep floors and benches clear; return tools to storage; clean spills promptly; provide ventilation; follow instructions; dispose of waste correctly; keep walkways unobstructed. (1 each)

13. Drawing and measurement

(a) Any THREE: title/name of object; dimensions; scale where applicable; labels/notes; material specification; date/drawing number; views or relevant details. (1 each)

(b) Dimensions communicate exact sizes and positions, enabling accurate marking out, cutting and assembly. (2)

(c) Any TWO: check the tool for damage; start at the correct zero mark; keep the ruler/tape straight; read at eye level; avoid stretching a tape measure; store safely. (1 each)

14. Energy, tools and product evaluation

(a) Renewable energy is replenished naturally on a human timescale; non-renewable energy comes from finite sources that are not quickly replaced. (1 each)

(b) Any TWO: clean and dry; sharpen where appropriate; oil moving parts; repair loose handles; store in a dry secure place. (1 each)

(c) Any THREE: function/performance; safety; strength/durability; accuracy of dimensions; quality of finish; appearance; cost; efficient use of materials; environmental impact. (1 each)

Teacher’s note: This is original teacher-prepared assessment material. Confirm that the relevant Grade 9 content and workshop procedures have been taught before administering. Accept other technically correct responses.